

Calculus II

Improper integrals

Todor Milev

2019

Outline

1 Improper Integrals

- Type I: Infinite Intervals
- Type II: Discontinuous Integrands
- A Comparison Test for Improper Integrals

License to use and redistribute

These lecture slides and their $\text{\LaTeX}$ source code are licensed to you under the Creative Commons license CC BY 3.0. You are free

- to Share - to copy, distribute and transmit the work,
- to Remix - to adapt, change, etc., the work,
- to make commercial use of the work,

as long as you reasonably acknowledge the original project.

- Latest version of the .tex sources of the slides:
<https://github.com/tmilev/freecalc>
- Should the link be outdated/moved, search for “freecalc project”.
- Creative Commons license CC BY 3.0:
<https://creativecommons.org/licenses/by/3.0/us/>
and the links therein.

Improper Integrals

- The definition of $\int_a^b f(x)dx$, where f is defined on $[a, b]$, has two requirements:
 - 1 $[a, b]$ is a finite interval.
 - 2 f has no infinite discontinuities in $[a, b]$.
- We are now going to relax these requirements.
 - 1 We allow infinite intervals, such as (a, ∞) , $(-\infty, b)$, and $(-\infty, \infty)$.
 - 2 f might have infinite discontinuities in $[a, b]$.
- Such integrals are called improper integrals.

Definition (Improper Integral)

The integral

$$\int_a^b f(x)dx$$

is called improper if one or more of the endpoints a and b is infinite, or if f has an infinite discontinuity on $[a, b]$.

Type I: Infinite Intervals

- Consider the region A that lies under $y = 1/x^2$, above the x -axis, and to the right of $x = 1$.
- To find its area, approximate with $A(t)$, the area of the region under $1/x^2$, above the x -axis, right of $x = 1$, and left of $x = t$.

$$A(t) = \int_1^t \frac{dx}{x^2} = \left[-\frac{1}{x} \right]_1^t = 1 - \frac{1}{t}$$

Definition (Improper Integral of Type I)

- ❶ If $\int_a^t f(x)dx$ exists for every $t \geq a$, then

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

if the limit exists.

- ❷ If $\int_t^b f(x)dx$ exists for every $t \leq b$, then

$$\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx$$

if the limit exists.

$\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called convergent if the corresponding limit exists and divergent if it doesn't exist.

- ❸ If both $\int_a^\infty f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define

$$\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^\infty f(x)dx.$$

Example

Determine whether $\int_1^{\infty} \frac{1}{x} dx$ is convergent or divergent.

Infinite area

Finite area

$$\begin{aligned}\int_1^{\infty} \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx \\ &= \lim_{t \rightarrow \infty} [\ln x]_1^t \\ &= \lim_{t \rightarrow \infty} (\ln t - \ln 1) \\ &= \lim_{t \rightarrow \infty} \ln t = \infty\end{aligned}$$

Therefore the improper integral is divergent.

Example

Evaluate

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx.$$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx \\ &= \frac{\pi}{2} + \frac{\pi}{2} = \pi \end{aligned}$$

Evaluate the two integrals separately:

$$\begin{aligned} \int_{-\infty}^0 \frac{1}{1+x^2} dx &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx = \lim_{t \rightarrow -\infty} [\arctan x]_t^0 \\ &= \lim_{t \rightarrow -\infty} (\arctan 0 - \arctan t) = \lim_{t \rightarrow -\infty} (0 - \arctan t) \\ &= 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \int_0^{\infty} \frac{1}{1+x^2} dx &= \lim_{t \rightarrow \infty} \int_0^t \frac{1}{1+x^2} dx = \lim_{t \rightarrow \infty} [\arctan x]_0^t \\ &= \lim_{t \rightarrow \infty} (\arctan t - \arctan 0) = \lim_{t \rightarrow \infty} \arctan t = \frac{\pi}{2} \end{aligned}$$

Example

For what values of p is the integral $\int_1^\infty \frac{1}{x^p} dx$ convergent?

- We know from Example 1 that if $p = 1$, the integral is divergent.
- Assume $p \neq 1$.

$$\int_1^\infty \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^t = \lim_{t \rightarrow \infty} \frac{\frac{1}{t^{p-1}} - 1}{1-p}$$

- If $p > 1$, then $p - 1 > 0$, so as $t \rightarrow \infty$, $t^{p-1} \rightarrow \infty$ and $1/t^{p-1} \rightarrow 0$.
- Therefore $\int_1^\infty \frac{1}{x^p} dx = \frac{1}{p-1}$ if $p > 1$, and so the integral is convergent.
- If $p < 1$, then $p - 1 < 0$, so $\frac{1}{t^{p-1}} = t^{1-p} \rightarrow \infty$ as $t \rightarrow \infty$.
- Therefore $\int_1^\infty \frac{1}{x^p} dx$ is divergent if $p < 1$.

Theorem

$\int_1^\infty \frac{1}{x^p} dx$ converges if $p > 1$ and diverges if $p \leq 1$.

Type II: Discontinuous Integrands

We can use the same approach if the function f is discontinuous at one of the endpoints a and b in the integral $\int_a^b f(x)dx$.

For example, $\frac{1}{\sqrt{x-2}}$ is discontinuous at 2, so we might wonder if the integral

$$\int_2^5 \frac{1}{\sqrt{x-2}} dx$$

exists.

Definition (Improper Integral of Type II)

- ① If f is continuous on $[a, b)$ and discontinuous at b , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

if the limit exists.

- ② If f is continuous on $(a, b]$ and discontinuous at a , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$

if the limit exists.

$\int_a^b f(x)dx$ is called convergent if the corresponding limit exists and divergent if it doesn't exist.

- ③ If f has a discontinuity at c , where $a < c < b$, and both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then we define

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Example

Find $\int_2^5 \frac{1}{\sqrt{x-2}} dx$.

Observe that $x = 2$ is a vertical asymptote for the integrand.

Area = $2\sqrt{3}$

$$\begin{aligned}\int_2^5 \frac{1}{\sqrt{x-2}} dx &= \lim_{t \rightarrow 2^+} \int_t^5 \frac{1}{\sqrt{x-2}} dx \\ &= \lim_{t \rightarrow 2^+} \left[2\sqrt{x-2} \right]_t^5 \\ &= \lim_{t \rightarrow 2^+} 2(\sqrt{5-2} - \sqrt{t-2}) \\ &= 2\sqrt{3}\end{aligned}$$

Example

Evaluate $\int_0^3 \frac{1}{x-1} dx$.

Observe that $x = 1$ is a vertical asymptote for the integrand.

$$\int_0^3 \frac{1}{x-1} dx = \int_0^1 \frac{1}{x-1} dx + \int_1^3 \frac{1}{x-1} dx$$

$$\begin{aligned} \int_0^1 \frac{dx}{x-1} &= \lim_{t \rightarrow 1^-} \int_0^t \frac{dx}{x-1} = \lim_{t \rightarrow 1^-} [\ln |x-1|]_0^t \\ &= \lim_{t \rightarrow 1^-} \ln |t-1| - \ln 1 = -\infty \end{aligned}$$

- Therefore the integral diverges.
- If we had not noticed the vertical asymptote, we might have made the following **mistake**:

$$\int_0^3 \frac{dx}{x-1} = [\ln |x-1|]_0^3 = \ln 2 - \ln 1 = \ln 2.$$

A Comparison Test for Improper Integrals

Sometimes it's impossible to find the exact value of an integral, but we still want to know if it's convergent or divergent. For such cases, we can sometimes use the following theorem.

Theorem (Comparison Theorem)

Suppose f and g are continuous and $f(x) \geq g(x) \geq 0$ for $x \geq a$.

- ❶ *If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent.*
- ❷ *If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.*

A similar theorem holds for Type II improper integrals.

Example

Show that $\int_0^{\infty} e^{-x^2} dx$ is convergent.

- The antiderivative of e^{-x^2} isn't an elementary function.
- If integral were $\int_0^{\infty} e^{-x} dx$, we'd have no problem integrating.
- Notice that $0 \leq e^{-x^2} \leq e^{-x}$ for $x \geq 1$ (because $-x^2 < -x$ for $x > 1$ and the exponent is an increasing function).
- Split $\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx$.
- On the RHS, first integral is proper - no effect on convergence.

$$\int_1^{\infty} e^{-x^2} dx \leq \int_1^{\infty} e^{-x} dx$$

By the Comparison Theorem,

$$\int_1^{\infty} e^{-x^2} dx \text{ converges} \Rightarrow$$

$$\int_0^{\infty} e^{-x^2} dx \text{ converges.}$$

$$= \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx$$

$$= \lim_{t \rightarrow \infty} [-e^{-x}]_1^t$$

$$= \lim_{t \rightarrow \infty} (-e^{-t} - (-e^{-1}))$$

$$= e^{-1}$$

Example

Is $\int_1^{\infty} \frac{1 + e^{-x}}{x} dx$ convergent or divergent?

- Notice that for $x \geq 1$ we have $\frac{1 + e^{-x}}{x} > \frac{1}{x}$.
- By a previously studied example, $\int_1^{\infty} \frac{dx}{x}$ is divergent.
- Therefore $\int_1^{\infty} \frac{1 + e^{-x}}{x} dx$ is divergent by the Comparison Theorem.

Draw a unit circle as above, let O, A be as indicated. Let P_2 be the point $(x_2, 1)$, P_1 be the point $(x_2 - \Delta, 1)$. By the Pythagorean theorem, $|OP_2|^2 = 1 + x_2^2$ and similarly $|OP_1|^2 = 1 + (x_2 - \Delta)^2$. Let Q_1, Q_2 be as indicated. Then $\triangle OP_2A$ is similar to $\triangle OAQ_2$. By Euclidean geometry,

$$\frac{|OA|}{|OP_2|} = \frac{|OQ_2|}{|OA|} \text{ and so } |OQ_2||OP_2| = |OA|^2 = 1 \text{ and therefore}$$

$$\frac{|OQ_2|}{|OP_2|} = \frac{|OQ_2||OP_2|}{|OP_2|^2} = \frac{1}{|OP_2|^2} = \frac{1}{1+x_2^2}. \text{ Similarly conclude}$$

$$|OQ_1||OP_1| = |OA|^2 = 1 = |OQ_2||OP_2|. \text{ Therefore } \frac{|OQ_1|}{|OP_2|} = \frac{|OQ_2|}{|OP_1|} \text{ and so}$$

$$\triangle OQ_2Q_1 \text{ is similar to } \triangle OP_1P_2. \text{ Therefore } \frac{|Q_1Q_2|}{|P_1P_2|} = \frac{|OQ_2|}{|OP_1|} \text{ and so}$$

$$|Q_1Q_2| = \frac{|P_1P_2||OQ_2|}{|OP_1|} = \left(\frac{|OP_2|}{|OP_1|}\right) \frac{|OQ_2|}{|OP_2|} |P_1P_2| = \frac{|OP_2|}{|OP_1|} \frac{\Delta}{1+x_2^2}.$$

$|Q_1 Q_2| = \frac{|OP_2|}{|OP_1|} \frac{\Delta}{1+x_2^2}$. For any $\varepsilon > 0$, can choose Δ : $1 < \frac{|OP_2|}{|OP_1|} < 1 + \varepsilon$.

If we let $P_2 \rightarrow P_1$, i.e., $\Delta \rightarrow 0$, we get $\frac{|OP_2|}{|OP_1|} \rightarrow 1$. In strict mathematical language: for every $\varepsilon > 0$ there exists $\delta > 0$ such that when $\Delta < \delta$ we have that $1 > \frac{|OP_2|}{|OP_1|} > 1 - \varepsilon$. Furthermore, the choice of δ can be made independent of the value of x_2 : to prove that one analyzes the

expression $\frac{|OP_2|}{|OP_1|} = \sqrt{\frac{1+x_2^2}{1+(x_2-\Delta)^2}}$. We leave the tedious but otherwise easy details to the interested student.

$|Q_1 Q_2| = \frac{|OP_2|}{|OP_1|} \frac{\Delta}{1+x_2^2}$. For any $\varepsilon > 0$, can choose Δ : $1 < \frac{|OP_2|}{|OP_1|} < 1 + \varepsilon$.

Fix a large number N and let Δ be such that $n = \frac{N}{\Delta}$ is integer. Let $P_0 = (0, 1)$, $P_1 = (\Delta, 1)$, $P_2 = (2\Delta, 1), \dots, P_n = (n\Delta, 1)$, and let $Q_0, Q_1, Q_2, \dots, Q_n$ be as indicated.

$$|Q_1 Q_2| = \frac{|OP_2|}{|OP_1|} \frac{\Delta}{1+x_2^2}. \text{ For any } \varepsilon > 0, \text{ can choose } \Delta: 1 < \frac{|OP_2|}{|OP_1|} < 1 + \varepsilon.$$

$$\frac{\Delta}{1+x_1^2} < |Q_0 Q_1| < (1 + \varepsilon) \frac{\Delta}{1+x_1^2}$$

$$\frac{\Delta}{1+x_2^2} < |Q_1 Q_2| < (1 + \varepsilon) \frac{\Delta}{1+x_2^2}$$

$$\vdots$$

$$\frac{\Delta}{1+x_n^2} < |Q_{n-1} Q_n| < (1 + \varepsilon) \frac{\Delta}{1+x_n^2}$$

$$\sum_{i=1}^n \frac{\Delta}{1+x_i^2} < \sum_{i=1}^n |Q_{i-1} Q_i| < (1 + \varepsilon) \sum_{i=1}^n \frac{\Delta}{1+x_i^2}$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \int_0^{\infty N} \frac{dx}{1+x^2} & = \lim_{\Delta, N, \varepsilon} \sum |Q_{i-1} Q_i| & = (1 + \varepsilon) \int_0^{\infty N} \frac{dx}{1+x^2} \end{array}$$

Let $\Delta \rightarrow 0$. Next take $N \rightarrow \infty$. Finally take $\varepsilon \rightarrow 0$, use squeeze thm.

$|Q_1 Q_2| = \frac{|OP_2|}{|OP_1|} \frac{\Delta}{1+x_2^2}$. For any $\varepsilon > 0$, can choose Δ : $1 < \frac{|OP_2|}{|OP_1|} < 1 + \varepsilon$.

The points $Q_1, Q_2, \dots$ see the segment OA from an angle of $\frac{\pi}{2}$.

Therefore, by Euclidean geometry, the points $Q_1, Q_2, \dots$ lie on the circle C with radius $\frac{1}{2}$ and center $(0, \frac{1}{2})$. Therefore $\sum |Q_{i-1} Q_i|$ approximates half of the circumference of the circle C . By symmetry,

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \text{circumference of } C = 2\pi \left(\frac{1}{2}\right) = \pi,$$

as desired.

We finish with illustration of the integral $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$. Recall

$$|Q_1 Q_2| \approx \frac{\Delta}{1+x_2^2}.$$